National Institute of Technology, Kurukshetra

B.Tech (5th Sem.)

Data Mining and Analytics (AIML)

Mid Sem. Exam- I

Duration: 50 min.

All questions are compulsory.

Max.Marks:15

1. Assume we are trying to mine the data, there could be two cases: (i) when we have billions of tuples to handle, that is, huge amount of data; and (ii) when we have few hundred tuples to process or mine. What could be the challenges that we would face with respect to the aforementioned cases? Explain in detail. **[05 Marks]**
2. Suppose you need to analyze the data set comprising the attribute named count. The tuples comprising this attribute include the following data values:
   46, 36, 35, 35, 15, 22, 33, 19, 13, 70, 45, 35, 52, 16, 25, 25, 22, 40, 33, 20, 16, 25, 35, 20, 30, 25, 21.
   Considering the given dataset, indicate the following observations:
   i. Mean
   ii. Median
   iii. Mode, also comment on the data's modality whether unimodal, bimodal or trimodal.
   iv. Mid-range.

   [03 Marks]
3. Differentiate between ratio-scaled and interval-scaled attribute. Which of the two falls under the category of discrete attributes? **[02Marks]**
4. How can we differentiate between K-means and K-medoid clustering algorithm? Which is preferred over other and why? Explain with the help of example. **[02 Marks]**
5. "For a given transactional database, we can fetch whether the given item set is frequent or not along with their actual support count by using either the close frequent item set or maximal frequent item set!", do you agree or disagree with the statement ? Justify your answer with the help of suitable example. **[03 Marks]**